Histology

Nasal Cavity

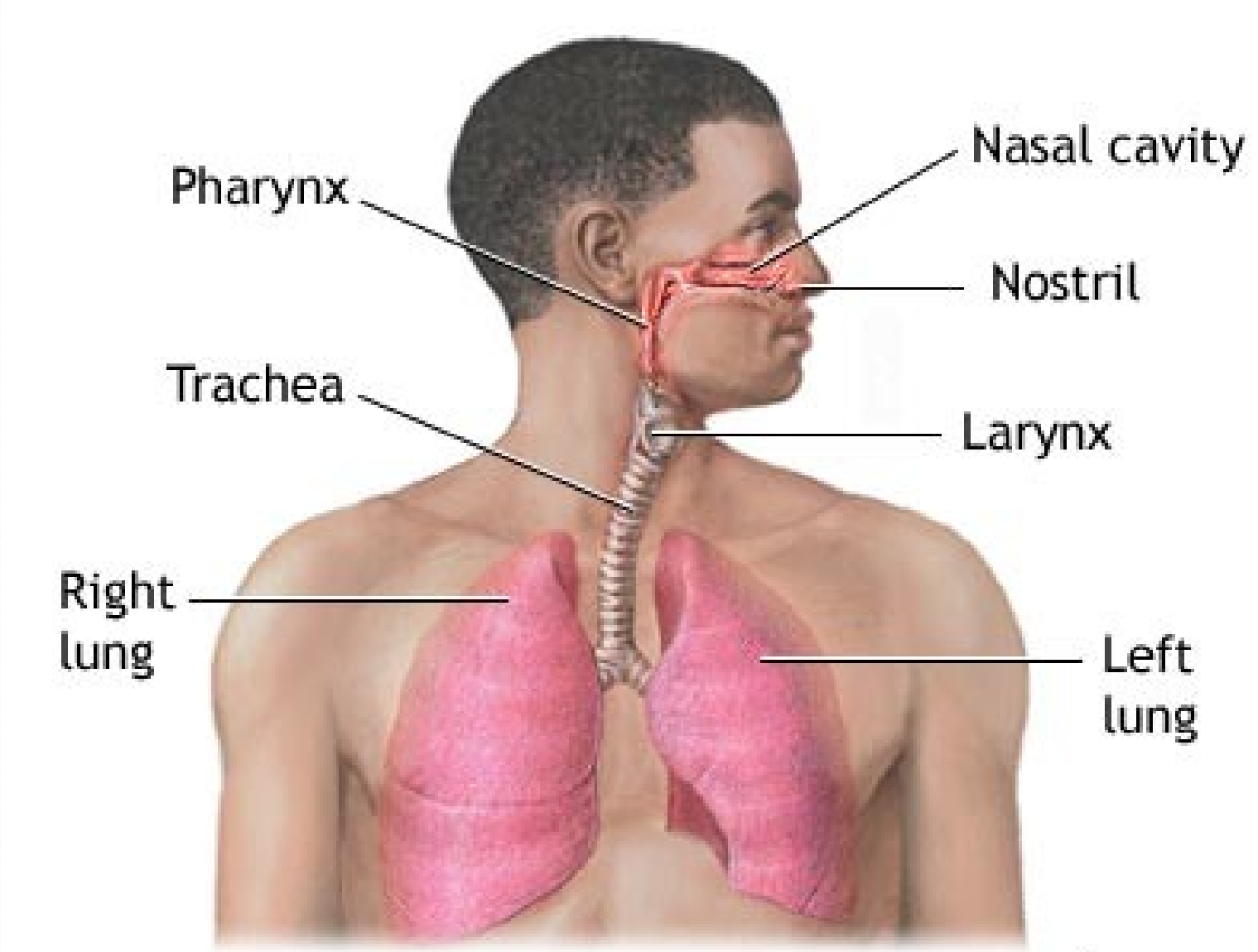
Respiratory & Olfactory Epithelium

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Learning objectives

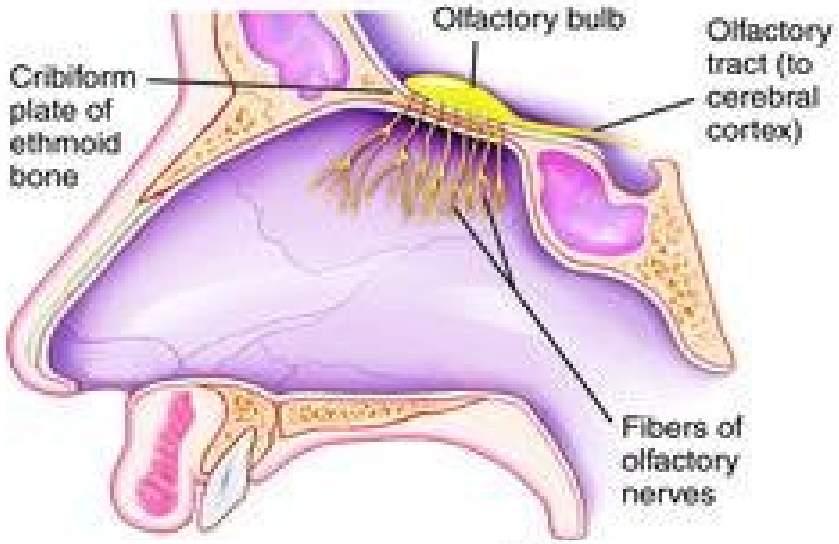
- At the end of the lecture the student should be able to describe:
- Details of respiratory epithelium
- Types of cells in respiratory epithelium.
- Olfactory epithelium
- Types of cells in olfactory epithelium.
- Mechanism of smell.
- Applied.

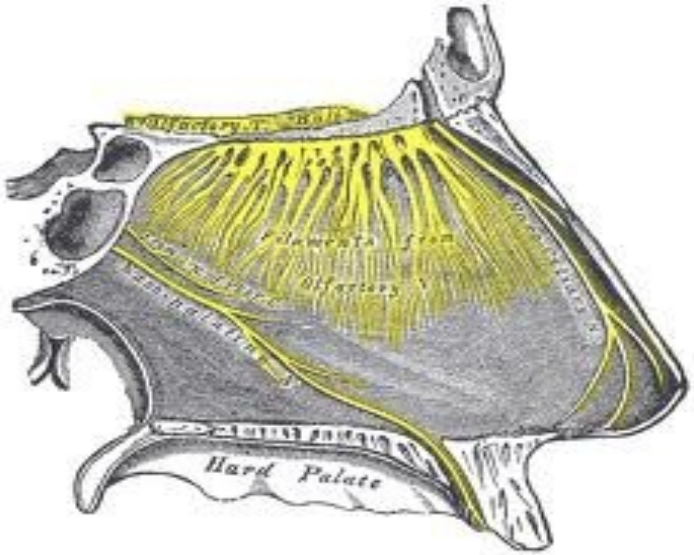
Respiratory system



Functions of Nose

- Warm incoming air.
- To trap some (by no means all) of the dirt and bacterial particles that are inhaled.
- The blood vessels underneath the epithelium serve to transfer heat to the incoming air flow, so that the humidity of the lung can be maintained at saturation



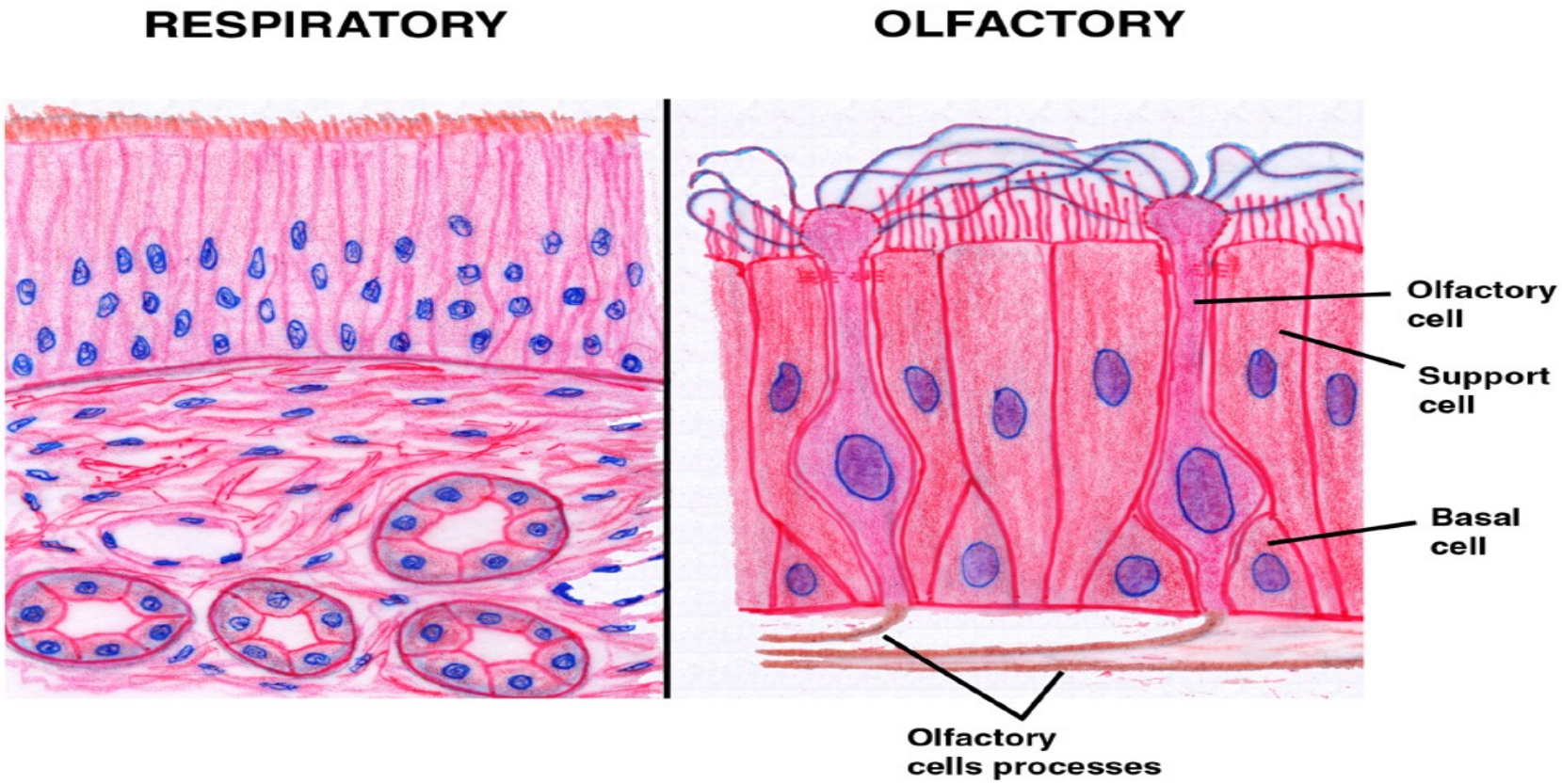


Epithelium of nasal cavity

The nasal cavity has two regions of epithelium,

1- Respiratory epithelium.

2- Olfactory epithelium which replaces respiratory epithelium in some regions.



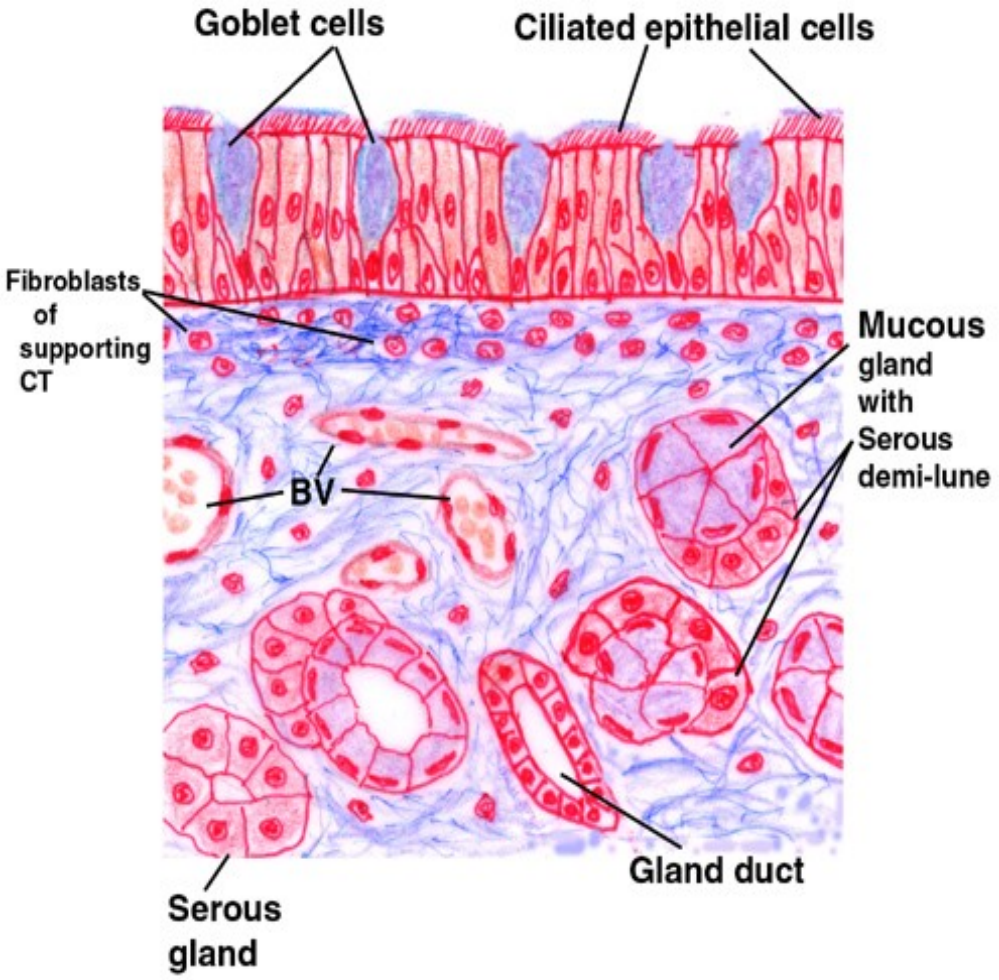
True respiratory epithelium

- There is a typical respiratory epithelium (TRE) throughout the length of air passages in the respiratory tract.
- Nose is the first place where TRE is present.



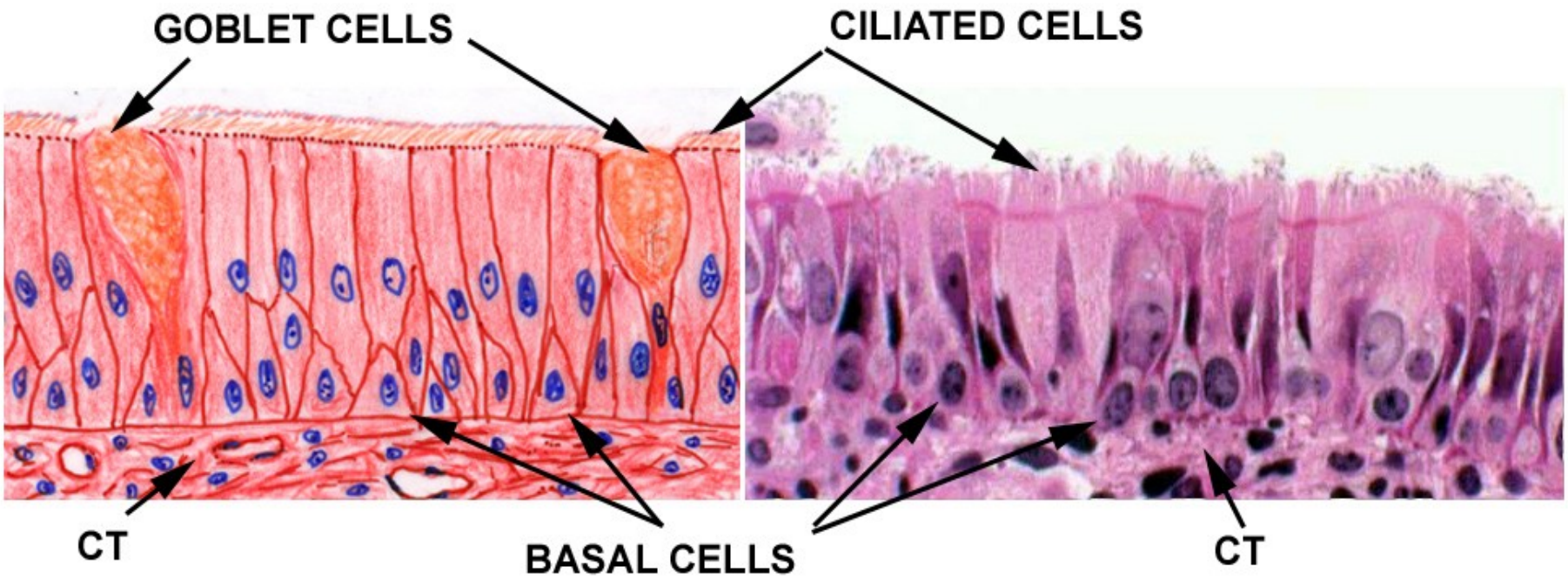
True Respiratory Epithelium

- TRE is a pseudostratified, ciliated, columnar epithelium in the upper airways.
- It rests on a CT lamina propria containing a large collection of glands and blood vessels.



Respiratory Epithelium

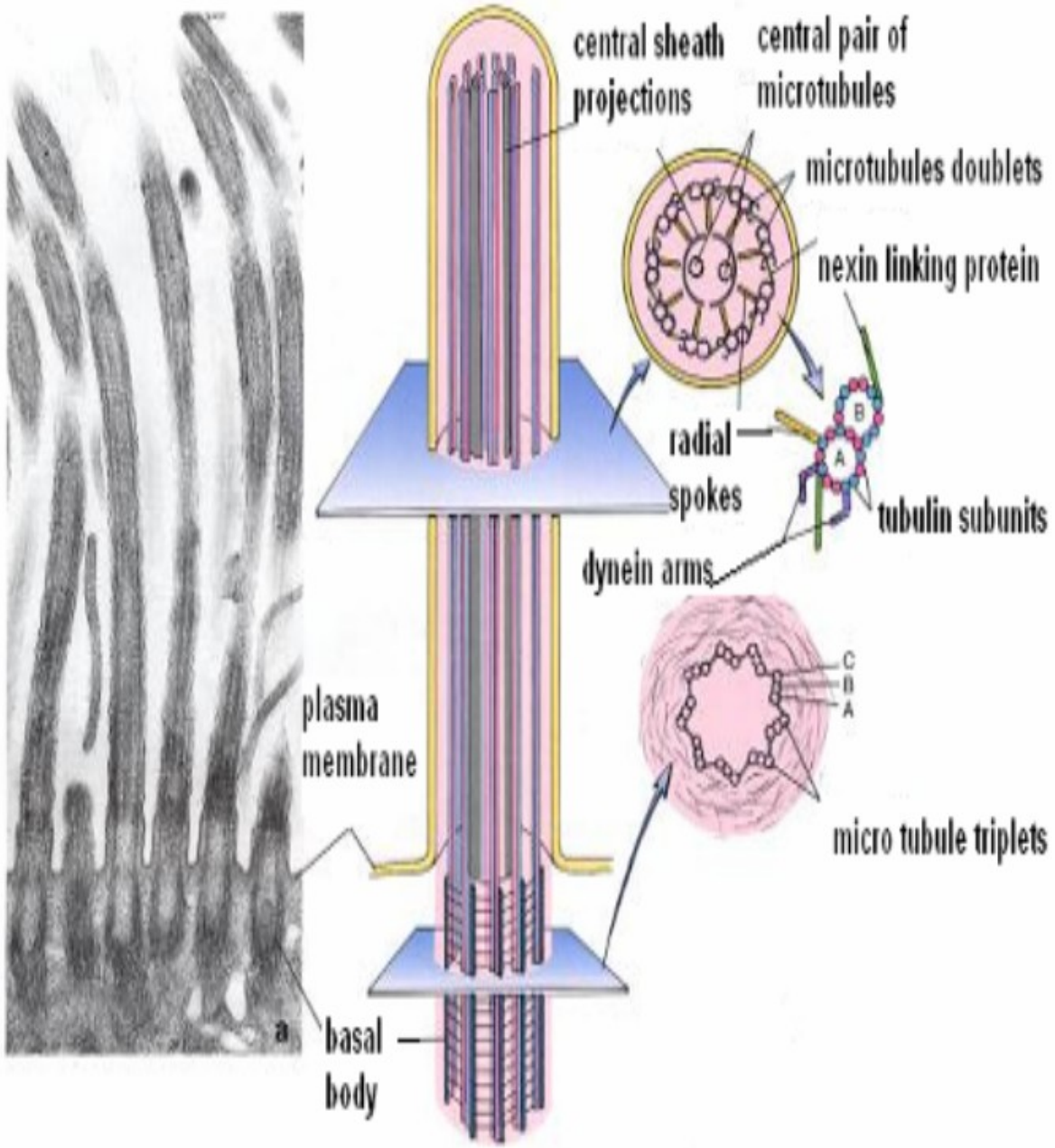
- Three main types of cells in upper tract
  - Ciliated
  - Goblet
  - Basal.



1- Ciliated columnar cells

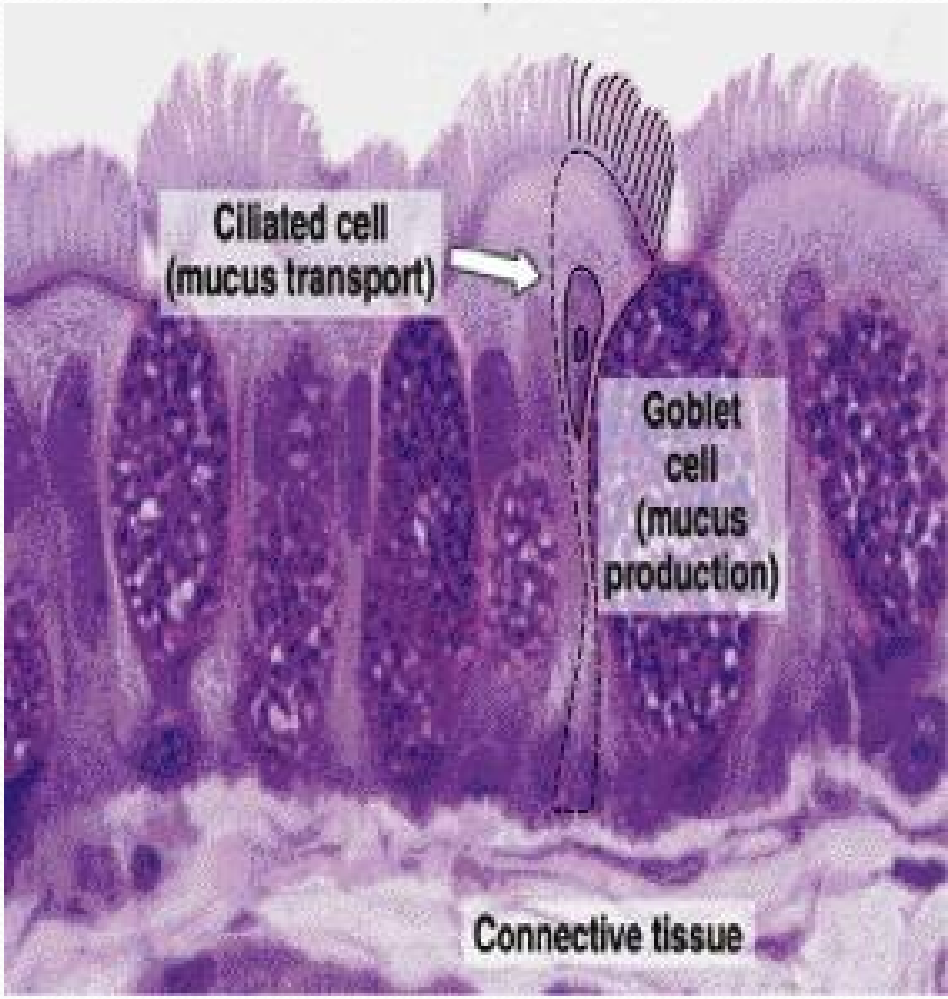
- Most abundant cell type
- Each cell has up to 300 ciliary tubules on the apical surface.
- Beneath the apical surface of the cell there are basal bodies and numerous mitochondria.
- Basal bodies serve as the anchoring sites for ciliary axonemes




central sheath projections
central pair of microtubules
microtubules doublets
nexin linking protein
radial spokes
tubulin subunits
dynein arms
plasma membrane
micro tubule triplets
basal body
a
B
A
C
B
A

2- Goblet cells

- Second most abundant cell type.
- Apical surface is packed with membrane bound, polysaccharide rich, mucous granules.
- Function is to secrete mucous to keep surfaces moist and to trap inhaled particles.

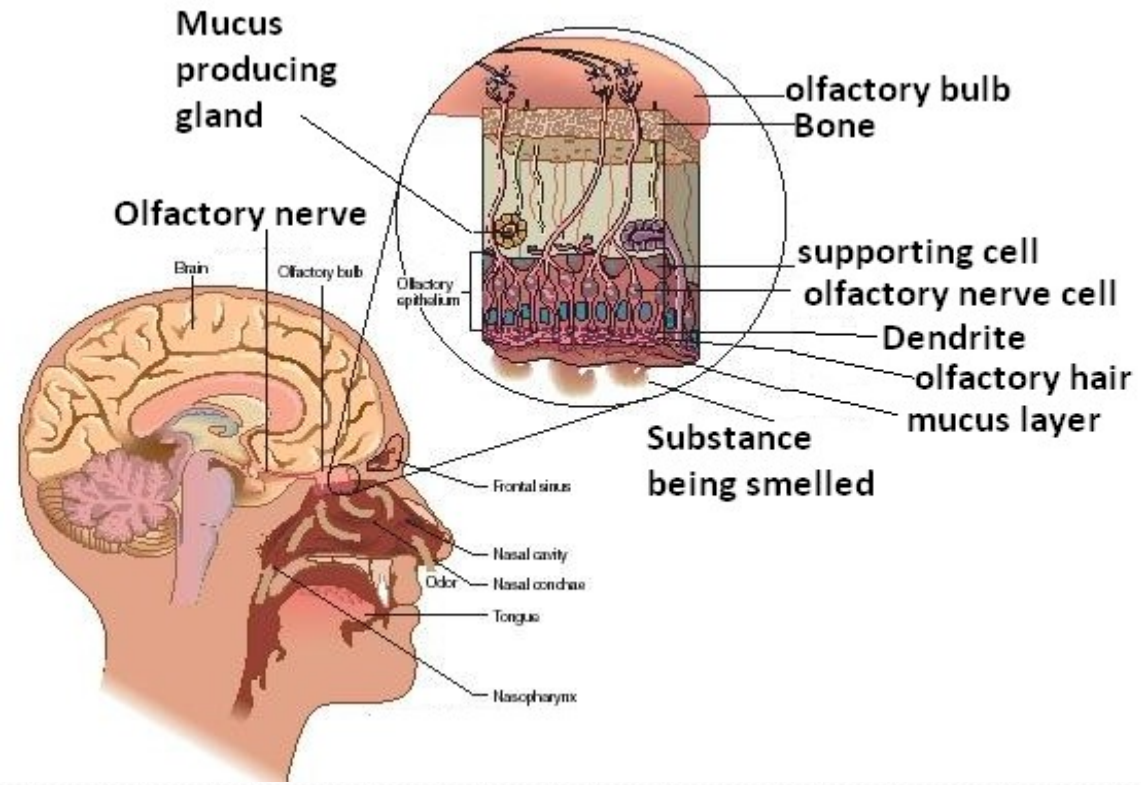


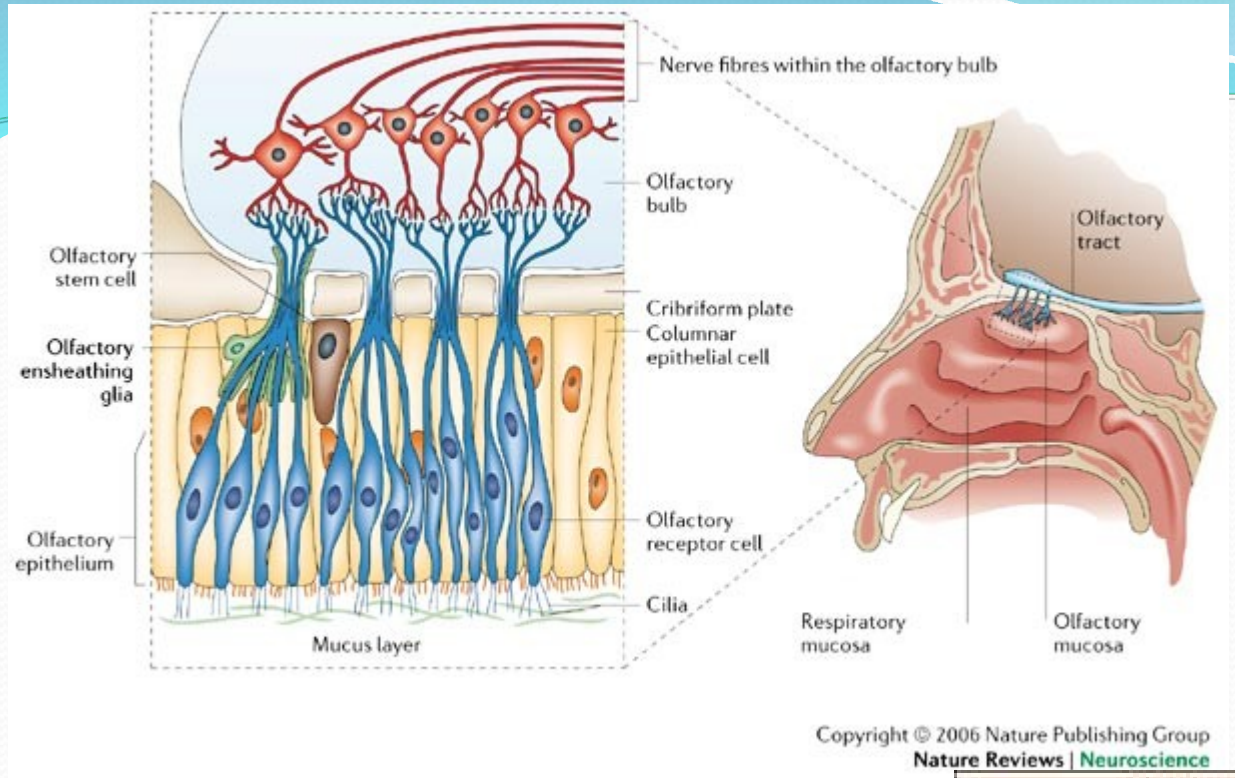
3- Basal (Short) Cells

- They are small cuboidal cells that lie on the basal lamina, but do not extend to the luminal surface of the epithelium.
- They are generative cells.
- They undergo mitosis and differentiate into other cell types.

Olfactory Epithelium

- **Olfactory epithelium** is a specialized epithelial tissue inside the nasal cavity that is involved in smell
- It is a part of olfactory system responsible for detecting odors.
- lies on the roof of the nasal cavity about 7 cm above and behind the nostrils





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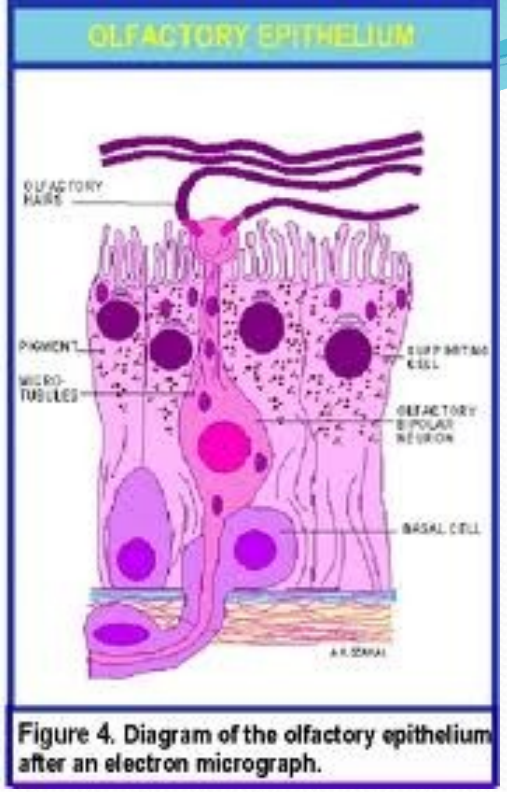
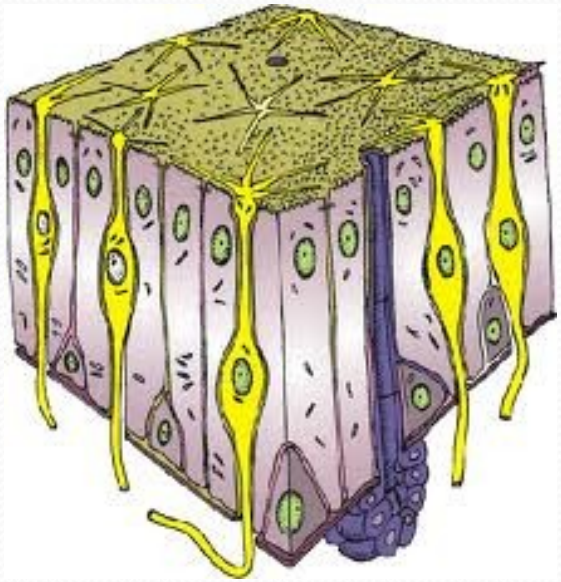
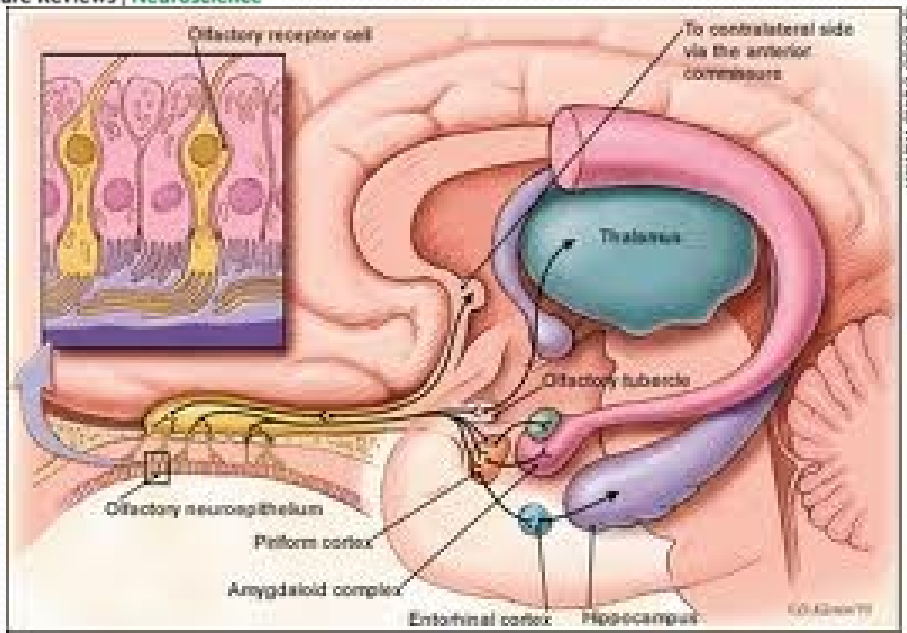


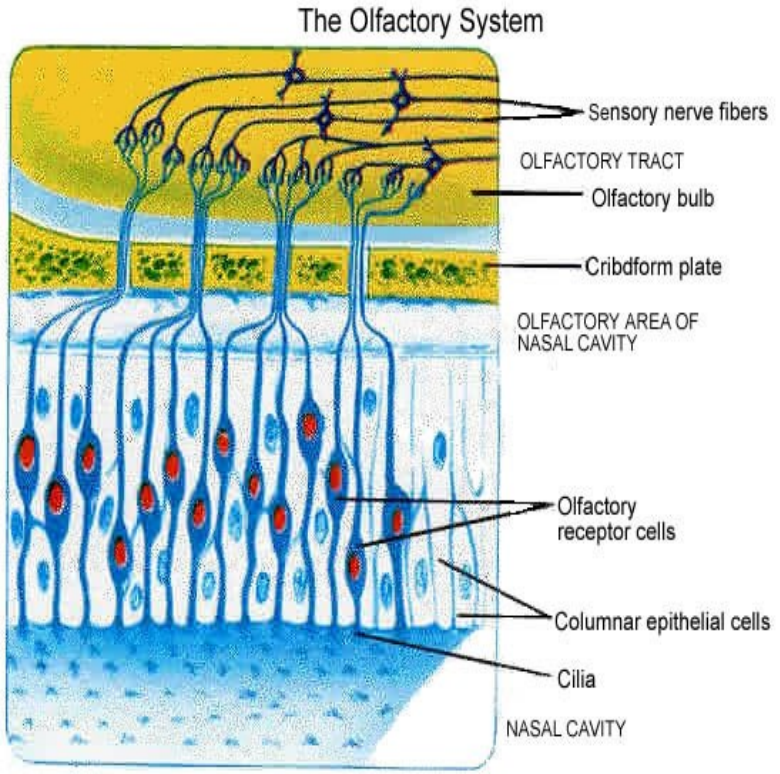
Figure 4. Diagram of the olfactory epithelium after an electron micrograph.

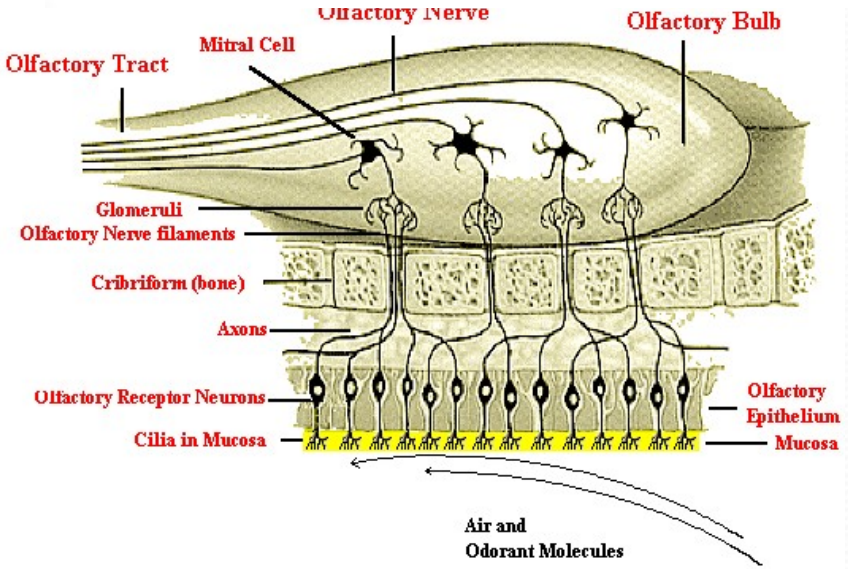




- Is also ciliated, but its function is not to trap dust and microbes, as is that of the TRE.
- It's a pseudostratified columnar type.
- The cilia are quite easily made out at the free surface but typically there are few or no goblet cells in the olfactory areas.

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Types of cells in olfactory epithelium

- Olfactory epithelium consists of three distinct types of cells
- 1. Olfactory cells
- 2. Supporting cells
- 3. Basal cells

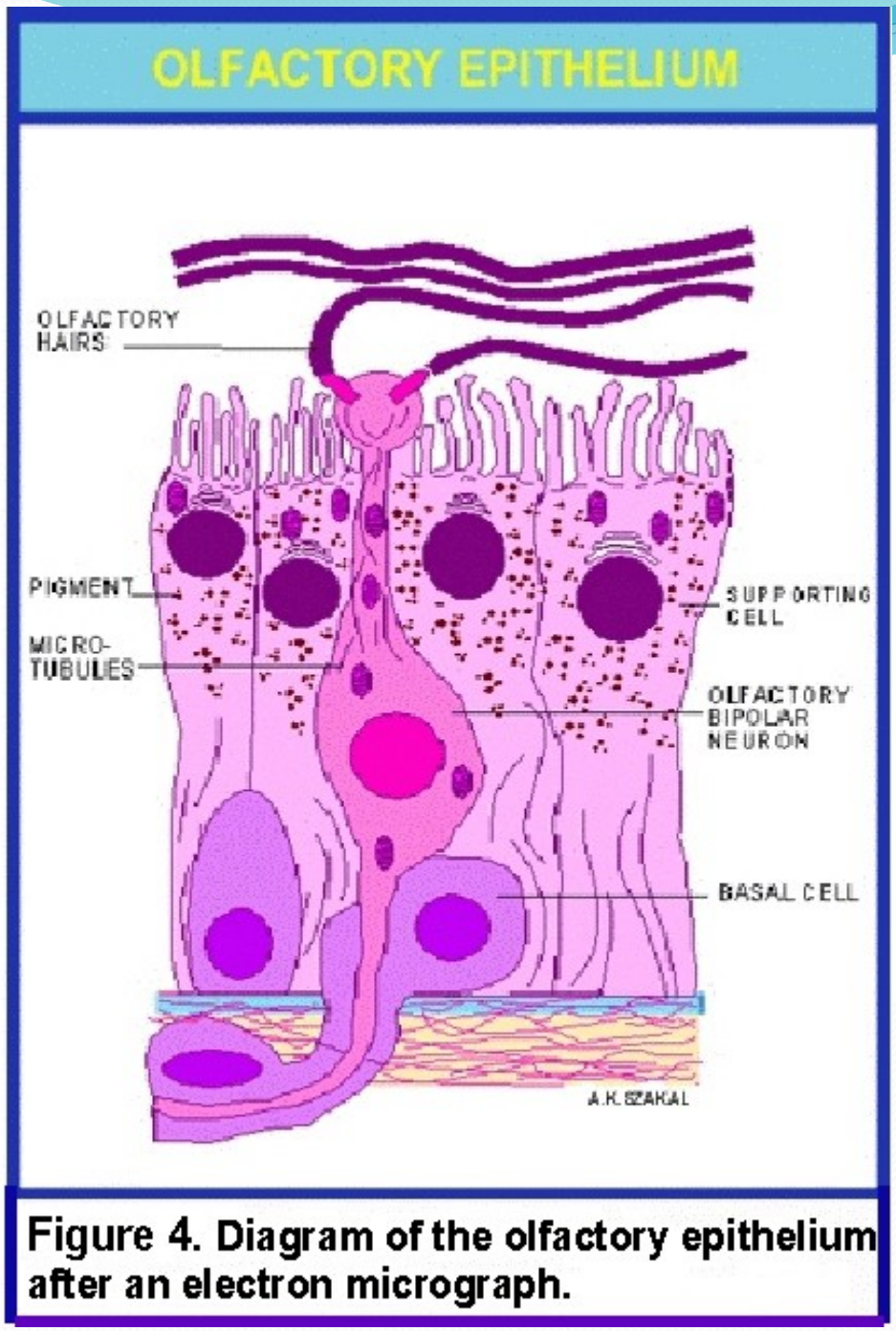
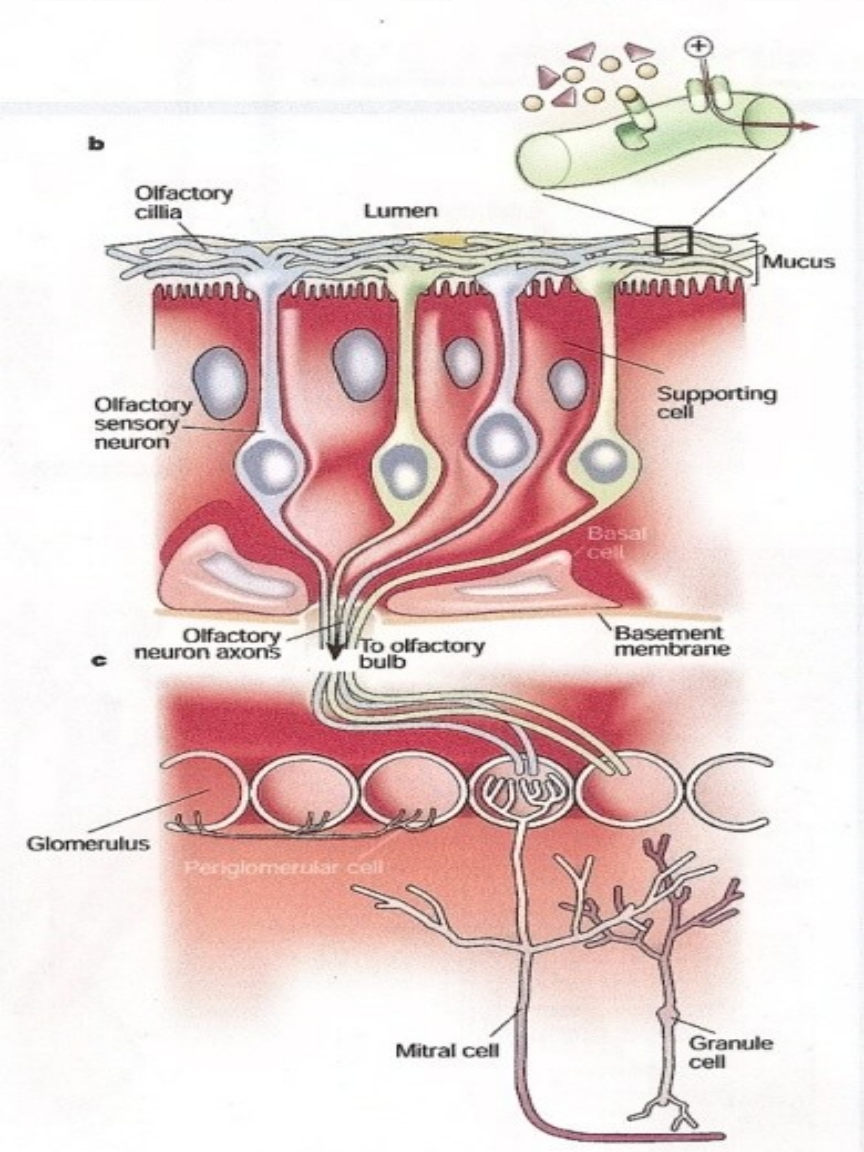


Figure 4. Diagram of the olfactory epithelium after an electron micrograph.

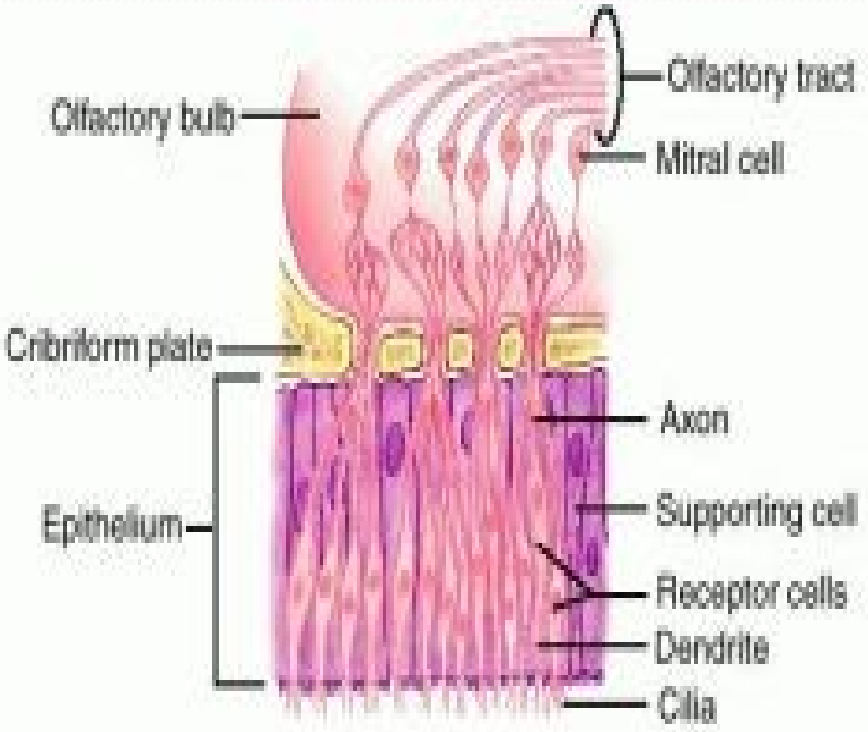
Olfactory cells

- Bipolar neurons which aggregate to form the olfactory nerve.
- The apical poles of these neurons are ciliated and coated with a serous secretion from Bowman's glands located in the lamina propria of the olfactory mucosa.



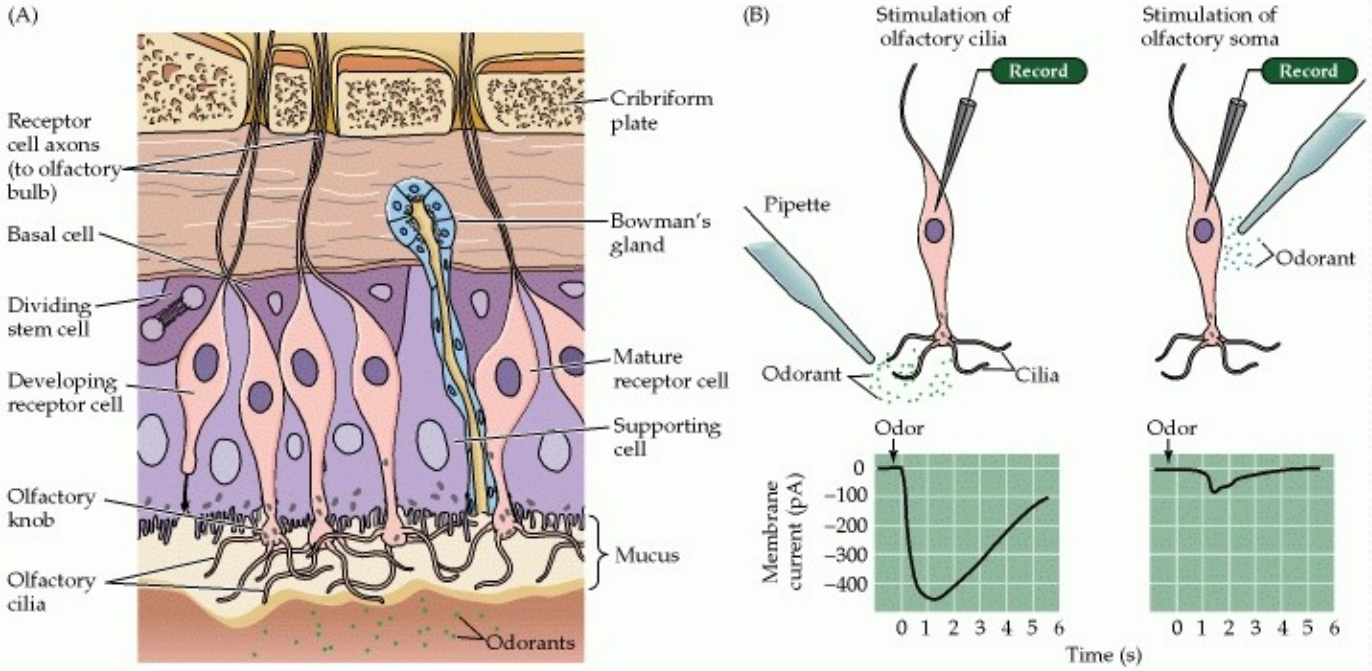
Supporting Cells

- Supporting cells (Sustentacular cells) of the olfactory epithelium functio as metabolic and physical support for the olfactory cells.
- Histologically, the supporting cells are tall columnar cells featuring microvilli and a prominent terminal web.
- The nuclei of supporting cells are more apically located than those of the other olfactory epithelial cells.



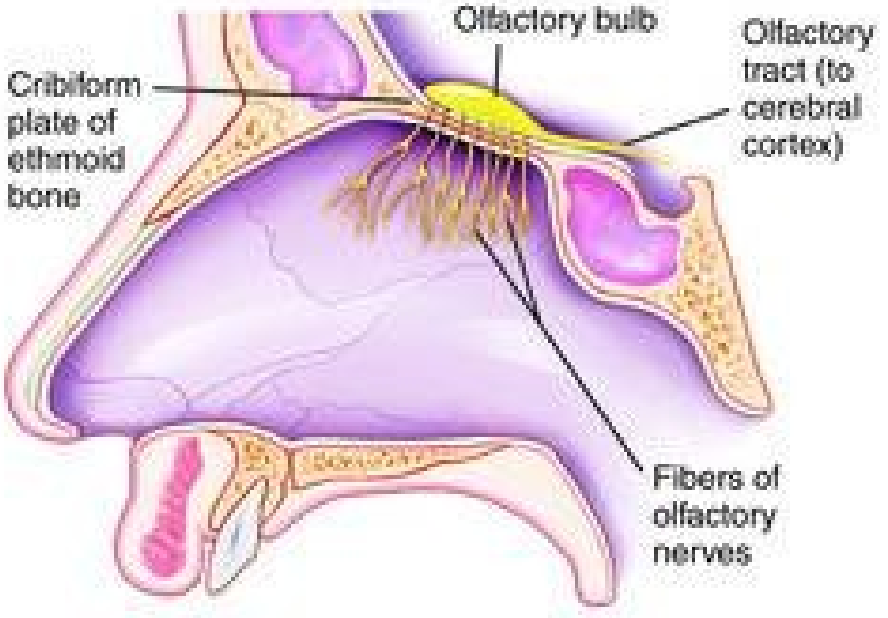
Basal cells

- Basal cells are stem cells capable of division and differentiation into either supporting or olfactory cells.
- The constant divisions of the basal cells leads to the olfactory epithelium being replaced every 2–4 weeks.
- Two types pf basal cells
  - Horizontal basal cells.
  - Superficial globose basal cells



Olfactory nerve

- The axons of the olfactory cells are bundled together to form the olfactory nerve.
- The olfactory nerve fibers pass through the bone of the nose into the olfactory lobe of the brain.

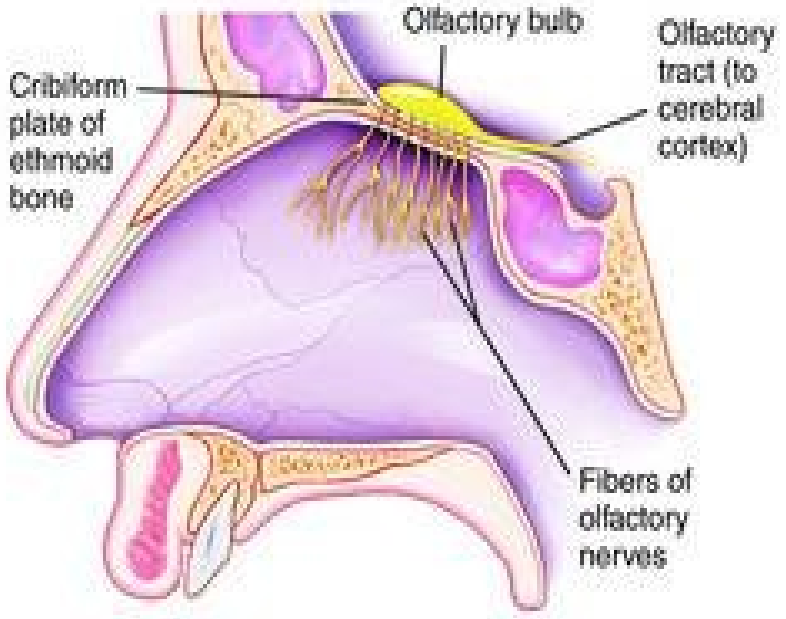


The Mechanism of Smell:

- The mechanism of smell is still very obscure.
- It is known that the plasma membrane of the olfactory cells is the actual site of chemoreception
- If the olfactory cells' binding sites aren't cleared of the materials bound to them, after a while they no longer transmit a signal, a phenomenon called "olfactory fatigue."
- After you've been in a barn for a few minutes you "don't smell it anymore."

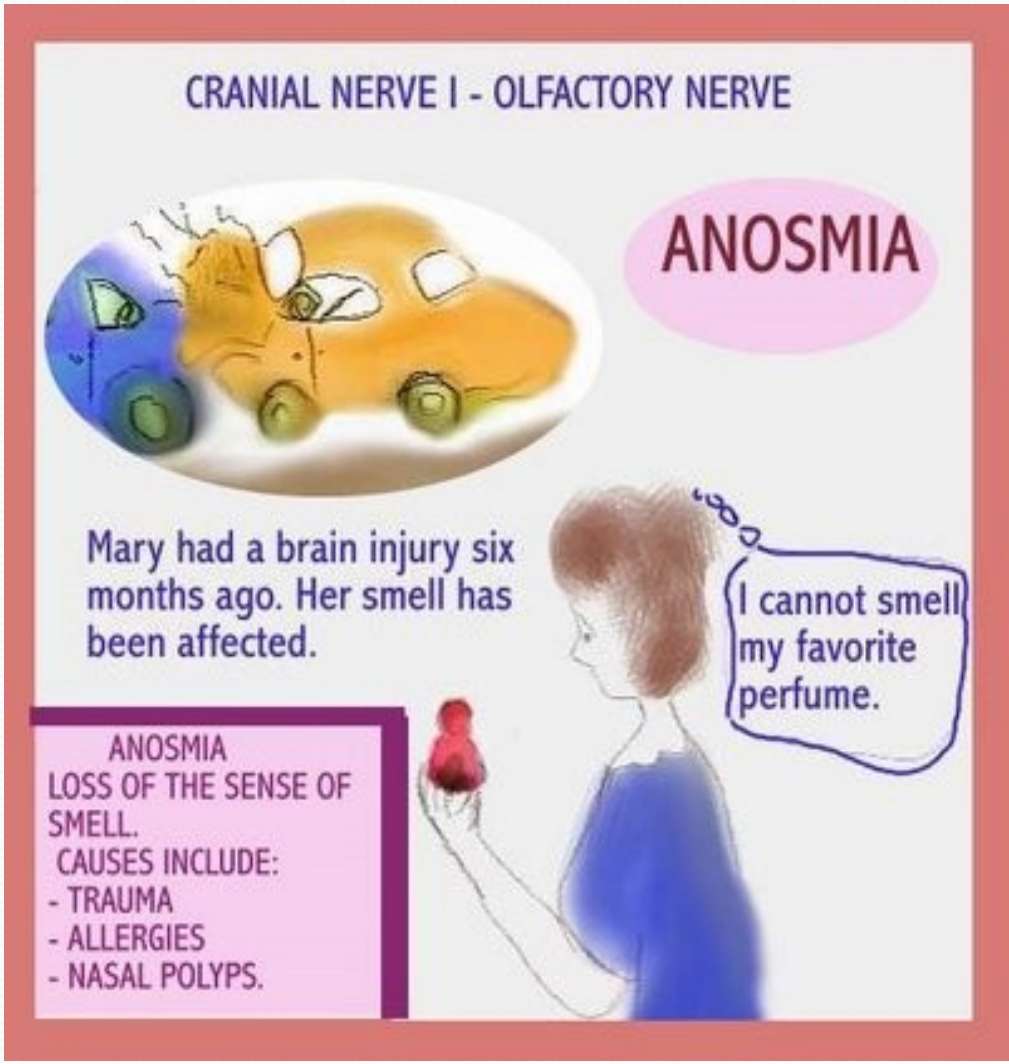
Functions of epithelium

- The serous and mucous secretions produced by the nasal glands underlying the epithelium (and the goblet cells embedded in the epithelial sheet) serve to catch and hold any dirt particles and bacteria that are inhaled.
- Keeping the surface moist provides a fluid sheet that traps things.
- The beating of the cilia being unidirectional.
- This is much less important in the nasal cavity than it is in the deeper parts of the respiratory tract, however.



Injury to olfactory epithelium

- The olfactory epithelium can be damaged by inhalation of toxic fumes, physical injury to the interior of the nose, and possibly by the use of some nasal sprays.
- Because of its regenerative capacity, damage to the olfactory epithelium can be temporary but in extreme cases, injury can be permanent, leading to anosmia.



Thank you